

SPRINT BACKLOG - Sprint 1 - SkyLens

Sprint Context

Duration: 3 weeks

Team: Product Owner, Scrum Master, 2 Mobile Developers, 1 UX/UI Designer

Team capacity: around 15-18 story points

Sprint Goal

Build and validate the technical foundation for real-time celestial object identification via camera (AR), delivering a functional end-to-end flow, from permission request to displaying the identified object on screen, reducing the project's biggest technical risk early in the first cycle.

Selected Items

All items below meet the project's **Definition of Ready**: clear "As a... I want... so that..." format, defined acceptance criteria, identified dependencies, team-estimated story points, and no known blockers.

[TASK] Integrate AR SDK with device sensors

Story points: 5 | **Suggested owner:** Mobile Developer 1

Complementary description:

Configure the AR SDK (ARKit for iOS / ARCore for Android) and connect it with the device's orientation sensors (gyroscope, magnetometer) and GPS, establishing the technical foundation to overlay information on the camera image in real time. This is a technical enabler with no direct end-user value on its own, but it unblocks all subsequent AR-related stories.

Acceptance Criteria:

- AR SDK integrated and running on a physical device (not only emulator)
 - App captures device orientation (azimuth and altitude) with $\pm 5^\circ$ precision
 - GPS data (latitude/longitude) is correctly captured for celestial position calculation
 - Tested on at least 1 Android and 1 iOS device
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[STORY] As Ana, I want to point my phone's camera to the sky and see the celestial object identified

Story points: 8 | **Suggested owner:** Mobile Developer 1 + Mobile Developer 2 (pair programming, given the technical risk)

Full user story: As Ana (casual user, curious about the night sky), I want to point my phone's camera to the sky and see the name of the celestial object overlaid on the screen, so that I can identify what I'm seeing without effort.

Complementary description:

Depends on "Integrate AR SDK with device sensors" being completed first. The calculation of which object is in the pointed direction must cross device orientation + GPS + a celestial position catalog (an open-source astronomical library may be used for ephemeris calculation in this first version).

Acceptance Criteria:

- Given the app has camera and GPS access, when I point to a visible celestial object, then its name appears overlaid within 2 seconds
 - If no object is recognized in the pointed direction, the app shows "No object identified here" instead of freezing or showing a blank screen
 - The overlaid text remains readable (adequate contrast) in both clear and dark sky
 - Tested with at least 3 different objects (e.g., Moon, Venus, a reference star)
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[STORY] As Ana, I want to grant camera and location permissions clearly

Story points: 2 | **Suggested owner:** UX/UI Designer + Mobile Developer 2

Full user story: As Ana (casual user, curious about the night sky), I want to grant camera and location permissions clearly, so that I understand why the app needs this access.

Complementary description:

A simple onboarding screen shown before first camera use, explaining the reason for each requested permission in plain language.

Acceptance Criteria:

- Permission screen explains in one sentence the reason for each permission (camera and location)
 - If permission is denied, the app guides the user on how to enable it later in settings
 - Flow tested in both scenarios: permission granted and permission denied
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Sprint Summary

Item	Type	Story Points
Integrat AR SDK with divide sensors.	Enabler (Task)	5
As Ana, I want to point my phone's camera to the sky and see the celestial object identified.	Story	8
As Ana, I want to grant camera and location permissions clearly.	Story	2
Total		15

Sprint Planning Notes

- Since "Integrate AR SDK" is a technical prerequisite for the camera identification story, the team should start both developers working in parallel: one advances the SDK integration while the other prepares the object-position matching logic (catalog + orientation data), integrating both once the enabler is ready
- The "grant permissions" story is independent from the other two items and can be developed in parallel by the UX/UI Designer + 1 developer, without creating a bottleneck
- The next-priority feature by WSJF ("View basic info about an identified object") was intentionally left out of this sprint, since it depends on the camera identification story being completed and validated first, respecting the principle of not starting work that depends on something not yet tested

DEFINITION OF READY (DoR)

An item is ready to enter a Sprint when:

- The story follows the "As a... I want... so that..." format and is clear to the whole team
- Acceptance criteria are defined and understood
- Technical dependencies (enablers) have been identified and, if possible, resolved beforehand
- The item has been estimated by the team (story points)
- There are no known external blockers (e.g., missing API, pending design decision)
- The team understands the value/motivation behind the item (not just the "what," but the "why")

DEFINITION OF DONE (DoD)

An item is considered complete when:

- Code implemented and reviewed (code review) by another team developer
- Acceptance criteria validated manually or through automated testing
- Feature tested on at least 2 different devices/screen sizes
- No known critical bugs or blockers
- Minimal documentation updated (if applicable)
- Build successfully generated, with no compilation errors

Non-Functional Requirements (part of the DoD)

- **Performance:** object identification responds within 2 seconds after pointing the camera
- **Usability:** main flow (open app → identify object) executable in a maximum of 3 taps
- **Reliability/Offline:** MVP features operate 100% without internet connection
- **Compatibility:** works on Android 10+ and iOS 15+
- **Resource consumption:** camera + sensor usage should not cause excessive device heating during continuous sessions of up to 15 minutes

[STORY] As Ana, I want to point my phone's camera to the sky and see the celestial object identified

SKYLENS-1 / SKYLENS-4

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A fazer

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Melhorar o História

Informações

Responsável

Não atribuído

Atribuir a mim

Pai

SKYLENS-1 MVP - Identify th

Data limite

Nenhum

Categorias

Nenhum

Team

Nenhum

Data de início

Nenhum

Sprint

SKYLENS Sprint 1

Story point estimate

8

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Descrição

[Feature: AR Sky Identification]

As Ana (*casual user, curious about the night sky*), I want to point my phone's camera to the sky and see the name of the celestial object overlaid on the screen, so that I can identify what I'm seeing without effort.

Depends on: "Integrate AR SDK with device sensors"

Acceptance Criteria:

Given the app has camera and GPS access, when I point to a visible celestial object, then its name appears overlaid within 2 seconds

If no object is recognized, the app shows "No object identified here" instead of freezing or showing a blank screen

The overlaid text remains readable (adequate contrast) in both clear and dark sky

[STORY] As Ana, I want to grant camera and location permissions clearly

SKYLENS-1 / SKYLENS-5

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SKYLENS-1 MVP - Identify th

Data limite

Nenhum

Categorias

Nenhum

Team

Nenhum

Data de início

Nenhum

Sprint

SKYLENS Sprint 1

Story point estimate

2

Features

feature-ar-identification

+

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Descrição

[Feature: AR Sky Identification]

As Ana (*casual user, curious about the night sky*), I want to grant camera and location permissions clearly, so that I understand why the app needs this access.

Acceptance Criteria:

Permission screen explains in one sentence the reason for each permission (camera and location)

If permission is denied, the app guides the user on how to enable it later in settings

Flow tested in both scenarios: permission granted and permission denied

Story point estimate: 2

Subtarefas

Adicionar subtarefa